Case Study: Adaptive Domain N-gram Benchmark

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NLP Course

Motivation

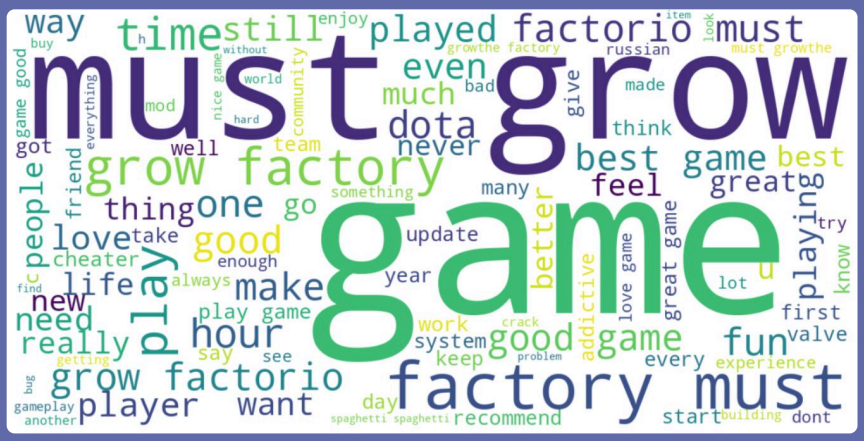
General-domain language models do not always transfer well to specialized domains. Gaming reviews contain domain-specific vocabulary and recurring phrase patterns, which motivates studying incremental adaptation of n-gram language models on in-domain data.

Experimental setup

Datasets
- Baseline corpus: Brown corpus (~140k tokens)
- Domain corpus: Steam reviews from Dota 2, CS and Factorio (~141k tokens)

Models
- 3-gram LM
- 5-gram LM
- Laplace smoothing

Evaluation metrics
- perplexity
- vocabulary coverage
- OOV rate



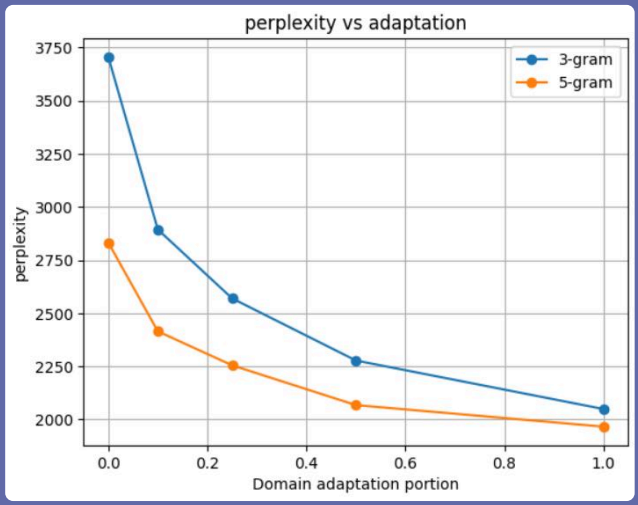
Methodology

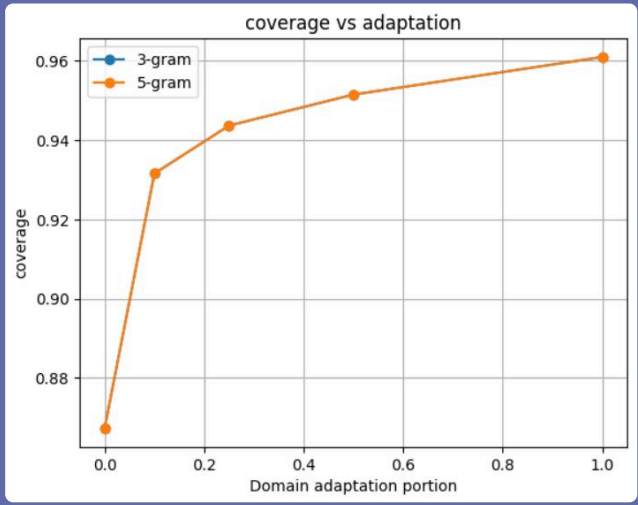
1. Train baseline 3-gram and 5-gram language models on the Brown corpus.
2. Incrementally adapt the models by adding 10%, 25%, 50%, and 100% of the gaming-domain training pool.
3. Evaluate each model on the same held-out gaming test set.
4. Compare perplexity, vocabulary coverage, OOV rate, and rare-word behavior.

Preprocessing: lowercasing, URL removal, punctuation removal, tokenization.
Stopwords were retained for language modeling and removed only for word cloud visualization.

Results

Adding gaming-domain data consistently improved in-domain performance.
- 3-gram perplexity: 3705 → 2049
- 5-gram perplexity: 2832 → 1967
- Coverage: 0.867 → 0.961
- OOV rate: 0.133 → 0.039



Conclusions

Incremental domain adaptation improves in-domain language modeling. Even small amounts of gaming text provide clear gains, while additional data further improves coverage and predictive quality. In this experiment, the 5-gram model consistently outperformed the 3-gram model.